

# Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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2026-03-19

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

## 1. Цели и задачи работы

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Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

## 2. Процесс выполнения лабораторной работы

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```
mvkalinichenko@mvkalinichenko:~$ ls /etc/ > file.txt
mvkalinichenko@mvkalinichenko:~$ ls >> file.txt
mvkalinichenko@mvkalinichenko:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
cifs-utils
```

Рисунок 1: Запись в файл

```
mvkalinichenko@mvkalinichenko:~$  
mvkalinichenko@mvkalinichenko:~$ grep .conf file.txt > conf.txt  
mvkalinichenko@mvkalinichenko:~$ cat conf.txt  
anthy-unicode.conf  
asound.conf  
brltty.conf  
chrony.conf  
cpupower-service.conf  
dconf  
dnsmasq.conf  
dracut.conf  
dracut.conf.d  
fprintd.conf  
fuse.conf  
host.conf  
idmapd.conf  
kdump.conf  
krb5.conf  
krb5.conf.d  
ld.so.conf  
ld.so.conf.d  
libaudit.conf  
libuser.conf
```

Рисунок 2: Поиск расширения .conf



```
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage01/report/pandoc/filters/pandocxnos/core.py
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage02/report/_resources/csl
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage02/report/bib/cite.bib
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage03/report/_resources/csl
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage03/report/bib/cite.bib
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage04/report/_resources/csl
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage04/report/bib/cite.bib
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage05/report/_resources/csl
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage05/report/bib/cite.bib
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage06/report/_resources/csl
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-per
sonal/stage06/report/bib/cite.bib
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentatio
n/report/_resources/csl
/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentatio
n/report/bib/cite.bib
/home/mvkalinichenko/git-extended/.git/hooks/commit-msg.sample
/home/mvkalinichenko/git-extended/.git/objects/c3
/home/mvkalinichenko/git-extended/.git/objects/cc
/home/mvkalinichenko/git-extended/.git/config
/home/mvkalinichenko/conf.txt
mvkalinichenko@mvkalinichenko:~$
```

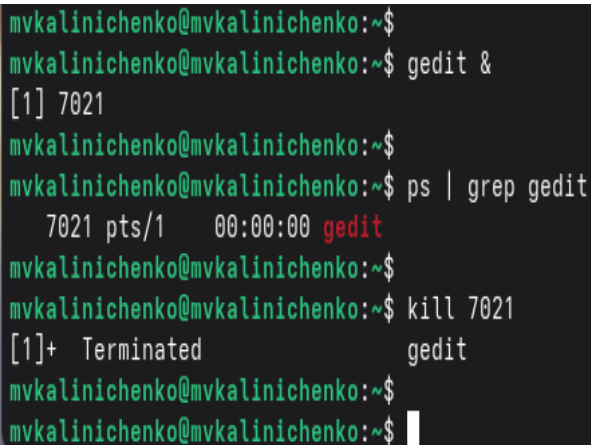
Рисунок 3: Поиск файлов

```
/etc/foomatic/hashes.d/hashes.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hpkip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
/etc/sysconfig/htcacheclean
/etc/systemd/system/httpd.service.d
/etc/systemd/system/multi-user.target.wants/happd.service
```

Рисунок 4: Поиск файлов

```
mvkalinichenko@mvkalinichenko:~$  
mvkalinichenko@mvkalinichenko:~$ find /etc -name "h*" -print | less  
find: '/etc/audit': Permission denied  
mvkalinichenko@mvkalinichenko:~$  
mvkalinichenko@mvkalinichenko:~$  
mvkalinichenko@mvkalinichenko:~$ find ~ -name "log*" > logfile &  
[1] 6983  
mvkalinichenko@mvkalinichenko:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
mvkalinichenko@mvkalinichenko:~$ rm logfile  
mvkalinichenko@mvkalinichenko:~$
```

Рисунок 5: Фоновый запуск процесса



```
mvkalinichenko@mvkalinichenko:~$  
mvkalinichenko@mvkalinichenko:~$ gedit &  
[1] 7021  
mvkalinichenko@mvkalinichenko:~$  
mvkalinichenko@mvkalinichenko:~$ ps | grep gedit  
    7021 pts/1    00:00:00 gedit  
mvkalinichenko@mvkalinichenko:~$  
mvkalinichenko@mvkalinichenko:~$ kill 7021  
[1]+  Terminated                  gedit  
mvkalinichenko@mvkalinichenko:~$  
mvkalinichenko@mvkalinichenko:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the file system containing each file name argument.  If no file name is given, the space available on all currently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows the space available on that file system rather than on the file system containing the device node.  This version of df cannot show the space available on unmounted file systems, because on most kinds of systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

    --direct
        show statistics for a file instead of mount point

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        produce a grand total

    -D, --dereference-args

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
mvkalinichenko@mvkalinichenko:~$  
mvkalinichenko@mvkalinichenko:~$ df  
Filesystem      1K-blocks    Used Available Use% Mounted on  
/dev/nvme0n1p3 155186176 90327244  64211204  59% /  
devtmpfs         4011288      0    4011288   0% /dev  
tmpfs            4035664      92   4035572   1% /dev/shm  
tmpfs           1614268    1952   1612316   1% /run  
tmpfs             1024        0      1024    0% /run/credentials/systemd-journald.service  
tmpfs           4035664    1612   4034052   1% /tmp  
/dev/nvme0n1p3 155186176 90327244  64211204  59% /home  
/dev/nvme0n1p2 1992552    518800   1352512  28% /boot  
/dev/loop2       49280     49280      0 100% /var/lib/snapd/snap/snapd/25935  
/dev/loop0       68480     68480      0 100% /var/lib/snapd/snap/core24/1349  
/dev/loop1      115712    115712      0 100% /var/lib/snapd/snap/hugo/25676  
tmpfs             1024        0      1024    0% /run/credentials/systemd-resolved.service  
tmpfs            807132      172   806960   1% /run/user/1045  
/dev/sr0        2677920    2677920      0 100% /run/media/mvkalinichenko/Fedora-WS-Live-43  
mvkalinichenko@mvkalinichenko:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/73
4      ./git-extended/.git/objects/cc
4      ./git-extended/.git/objects/2b
4      ./git-extended/.git/objects/1d
4      ./git-extended/.git/objects/b8
4      ./git-extended/.git/objects/f7
4      ./git-extended/.git/objects/bf
4      ./git-extended/.git/objects/9d
4      ./git-extended/.git/objects/61
4      ./git-extended/.git/objects/a7
4      ./git-extended/.git/objects/55
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
836952 .
mvkalinichenko@mvkalinichenko:~$
```

Рисунок 10: Запуск команды du



```
/home/mvkalinichenko/git-extended/.git/objects/74
/home/mvkalinichenko/git-extended/.git/objects/c3
/home/mvkalinichenko/git-extended/.git/objects/2a
/home/mvkalinichenko/git-extended/.git/objects/73
/home/mvkalinichenko/git-extended/.git/objects/cc
/home/mvkalinichenko/git-extended/.git/objects/2b
/home/mvkalinichenko/git-extended/.git/objects/1d
/home/mvkalinichenko/git-extended/.git/objects/b8
/home/mvkalinichenko/git-extended/.git/objects/f7
/home/mvkalinichenko/git-extended/.git/objects/bf
/home/mvkalinichenko/git-extended/.git/objects/9d
/home/mvkalinichenko/git-extended/.git/objects/61
/home/mvkalinichenko/git-extended/.git/objects/a7
/home/mvkalinichenko/git-extended/.git/objects/55
/home/mvkalinichenko/git-extended/.git/logs
/home/mvkalinichenko/git-extended/.git/logs/refs
/home/mvkalinichenko/git-extended/.git/logs/refs/heads
/home/mvkalinichenko/git-extended/.git/logs/refs/remotes
/home/mvkalinichenko/git-extended/.git/logs/refs/remotes/origin
/home/mvkalinichenko/monthly
/home/mvkalinichenko/reports
/home/mvkalinichenko/reports/monthly
/home/mvkalinichenko/reports/monthly/monthly
/home/mvkalinichenko/ski.places
/home/mvkalinichenko/ski.places/equipment
/home/mvkalinichenko/ski.places/plans
/home/mvkalinichenko/australia
/home/mvkalinichenko/play
/home/mvkalinichenko/play/games
/home/mvkalinichenko/play/games/play
mvkalinichenko@mvkalinichenko:~$
```

Рисунок 11: Поиск директорий

### 3. Выводы по проделанной работе

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В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.